

Improve metric distortion constant for deterministic best-of-both-worlds voting with full rankings

ChatGPT (gpt-5.6-sol)*

Solution generated on 2026-08-24

Abstract

For every fixed $0 < \alpha < 1$, we construct a deterministic ordinal social choice rule whose metric distortion is at most

$$3 + \frac{2\alpha}{1 - \alpha}$$

and whose utilitarian distortion is at most m^2/α , where m denotes the number of alternatives. For fixed α , the constants in these bounds are independent of the numbers of alternatives and voters. In particular, $\alpha = 1/3$ yields metric distortion at most 4 and utilitarian distortion at most $3m^2$; the target requirement of metric distortion below $9 + \delta$ for every $\delta > 0$, together with $O(m^2)$ utilitarian distortion, is thus completely resolved. More generally, for every fixed $\varepsilon > 0$, choosing $\alpha = \varepsilon/(2 + \varepsilon)$ yields metric distortion at most $3 + \varepsilon$ and utilitarian distortion at most $(1 + 2/\varepsilon)m^2$. The paper therefore presents a complete solution to the target problem, not merely partial progress.

1 The open problem

This paper addresses an open problem posed by Vasilis Gkatzelis, Mohamad Latifian, Nisarg Shah in *Best of Both Distortion Worlds* [1] (Economics and Computation (EC), 2023), Section 3 (Deterministic Voting Rule), discussion immediately after Theorem 2; page 8 of 21 in the provided text.. The website catalogue entry is problem 7 (W4383533377_p0) of the [Open Problems in Operations Research](#) collection.

Find a deterministic voting rule on full rankings whose utilitarian distortion is on the order of m^2 and whose metric distortion guarantee is strictly smaller than the currently known constant $9 + \varepsilon$ (for fixed $\varepsilon > 0$), i.e., improve the constant-factor upper bound on the worst-case metric distortion achievable simultaneously with an $\Theta(m^2)$ -order utilitarian distortion guarantee.

The remainder of this document is the solution produced by the model, reproduced verbatim from the pipeline output.

*Generated by the `base_solver_ni_two_iterations` pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve $\rightarrow$ A.8 verify $\rightarrow$ A.9 write-up; run `run_20260824_024352_21c1db8d, ec-2it-ultra`); two-iteration run, outcome from iteration 1. Independent verifier assessment: APPROVED with progress score 3/3 (full solution). Maintained by the AI in Mathematics & Operations Research project.

2 Model and main result

Let N be a finite set of $n \geq 1$ voters and let A be a finite set of $m \geq 1$ alternatives. Each voter $i \in N$ reports a strict total order $\succ_i$ on A , and we write $x \succeq_i y$ when either $x \succ_i y$ or $x = y$. A preference profile is $\sigma = (\succ_i)_{i \in N}$. A deterministic ordinal social choice rule maps every such profile to a single alternative and uses no cardinal utility or metric information.

A normalized utility profile consistent with σ consists of functions

$$u_i : A \longrightarrow \mathbb{R}_{\geq 0}, \quad \sum_{x \in A} u_i(x) = 1,$$

such that, for every $i \in N$ and all distinct $x, y \in A$,

$$x \succ_i y \implies u_i(x) > u_i(y).$$

The social welfare of $x \in A$ is

$$\text{SW}_{\mathbf{u}}(x) = \sum_{i \in N} u_i(x).$$

For a rule f , its utilitarian distortion at the given values of m and n is

$$\text{dist}_U(f) = \sup_{\sigma} \sup_{\mathbf{u} \text{ consistent with } \sigma} \frac{\max_{x \in A} \text{SW}_{\mathbf{u}}(x)}{\text{SW}_{\mathbf{u}}(f(\sigma))},$$

with the usual extended-real interpretation if the denominator is zero. The rule constructed below always has strictly positive selected welfare, so no zero-denominator convention is needed for its utilitarian bound.

An admissible metric is a function

$$d : (N \cup A) \times (N \cup A) \longrightarrow \mathbb{R}_{\geq 0}$$

that is symmetric, satisfies $d(z, z) = 0$, and obeys the triangle inequality. As is standard in metric social choice, distinct points may have distance zero; formally, the admissible metrics are therefore pseudometrics. The metric d is consistent with the profile if

$$x \succeq_i y \implies d(i, x) \leq d(i, y)$$

for every voter i and all alternatives x, y . The social cost of an alternative x is

$$\text{SC}_d(x) = \sum_{i \in N} d(i, x).$$

For metric distortion, a ratio with both numerator and denominator equal to zero is assigned the value 1, while a positive numerator divided by zero has the value $+\infty$. Under this convention,

$$\text{dist}_M(f) = \sup_{\sigma} \sup_{d \text{ consistent with } \sigma} \frac{\text{SC}_d(f(\sigma))}{\min_{x \in A} \text{SC}_d(x)}.$$

Equivalently, $\text{dist}_M(f) \leq C$ means that the undivided inequality

$$\text{SC}_d(f(\sigma)) \leq C \min_{x \in A} \text{SC}_d(x)$$

holds for every profile and every consistent metric. This formulation also covers instances in which the optimal social cost is zero.

Our main result is the following.

Theorem 1. *For every fixed constant $0 < \alpha < 1$, every $m \geq 1$, and every $n \geq 1$, there is a deterministic ordinal rule f^α , using the voters' full rankings, such that*

$$\text{dist}_M(f^\alpha) \leq 3 + \frac{2\alpha}{1 - \alpha}$$

and

$$\text{dist}_U(f^\alpha) \leq \frac{m^2}{\alpha}.$$

For fixed α , the constants in these bounds are independent of m and n .

3 Construction of the rule

Fix $0 < \alpha < 1$. For each alternative $x \in A$, let

$$p(x) = |\{i \in N : x \text{ is ranked first by } i\}|$$

denote its plurality score in the profile, and define the set of retained alternatives

$$H = \left\{ x \in A : p(x) \geq \frac{\alpha n}{m} \right\}.$$

This set is nonempty. Indeed,

$$\sum_{x \in A} p(x) = n,$$

so the finite averaging principle gives an alternative x with $p(x) \geq n/m$. Since $\alpha < 1$, this alternative satisfies $p(x) > \alpha n/m$ and therefore belongs to H .

For every voter i , let r_i denote the unique highest-ranked alternative of i among the members of H . For each $a \in H$, define a bipartite graph $D_H(a)$ as follows: both sides of the graph are copies of N , and there is an edge from the left copy of i to the right copy of j precisely when

$$a \succeq_i r_j.$$

A perfect matching in this graph is thus the same as a bijection $\pi : N \rightarrow N$ satisfying

$$a \succeq_i r_{\pi(i)} \quad \text{for every } i \in N.$$

Fix, before the profile is reported, a deterministic linear tie-breaking order on A . The rule f^α selects the first alternative in this fixed order among all $a \in H$ for which $D_H(a)$ has a perfect matching. The following lemma shows that at least one such alternative always exists.

Lemma 1. *For every preference profile, there is an alternative $a \in H$ such that $D_H(a)$ has a perfect matching.*

Proof. Place one token labeled j on the alternative r_j for every $j \in N$. This gives exactly n labeled tokens, all located on alternatives in H .

Process the n voters once each, in any fixed order. Immediately before a voter is processed, call an alternative *active* if it still holds at least one token. At least one alternative is active at that moment, because fewer than n tokens have been removed so far. The voter removes one token from their least-preferred active alternative; if that alternative holds more than one token, a fixed ordering of the token labels determines which token is removed. Consequently,

each voter removes exactly one token, and after all voters have been processed, all n tokens have been removed.

Let a be the alternative from which the last token was removed. Tokens never move, and every token was initially placed on some $r_j \in H$, so $a \in H$. Moreover, the last token remains on a until the final removal, so a is active immediately before every removal, including the final one.

For each voter i , let $\pi(i)$ be the label of the token removed by i ; that token is located on $r_{\pi(i)}$. Since $r_{\pi(i)}$ is voter i 's least-preferred active alternative at that step, while a is also active, it follows that

$$a \succeq_i r_{\pi(i)}.$$

Every voter removes exactly one token, every token is removed exactly once, and the token labels are precisely the members of N ; hence $\pi : N \rightarrow N$ is a bijection. The pairs $(i, \pi(i))$ are all edges of $D_H(a)$, so they form a perfect matching in that graph. $\square$

Lemma 1 shows that the eligible set used by f^α is nonempty, and the fixed tie-breaking order then makes the rule deterministic. Every ingredient of the construction—the plurality scores, the retained set H , the alternatives r_i , and the graphs $D_H(a)$ —is determined solely by the reported full rankings.

4 Utilitarian distortion

Lemma 2. *For every profile and every normalized utility profile consistent with it, the alternative $a = f^\alpha(\sigma)$ satisfies*

$$\text{SW}_{\mathbf{u}}(a) \geq \frac{\alpha n}{m^2}.$$

Consequently,

$$\text{dist}_U(f^\alpha) \leq \frac{m^2}{\alpha}.$$

Proof. The selected alternative belongs to H , so

$$p(a) \geq \frac{\alpha n}{m}.$$

Consider any voter i who ranks a first. The value $u_i(a)$ is the maximum of the m nonnegative numbers $\{u_i(x) : x \in A\}$, whose sum is 1; therefore

$$u_i(a) \geq \frac{1}{m}.$$

This conclusion is also valid under the stipulated strict consistency of the utilities: when $m > 1$, strict consistency in fact makes the top alternative's utility strictly greater than the average whenever equality of all m utilities would otherwise be required, but only the displayed weak inequality is needed.

Summing over the $p(a)$ first-place supporters of a , and using nonnegativity for all remaining voters, gives

$$\text{SW}_{\mathbf{u}}(a) = \sum_{i \in N} u_i(a) \geq \frac{p(a)}{m} \geq \frac{\alpha n}{m^2}.$$

In particular, the selected welfare is positive.

For every voter i and every alternative x , nonnegativity and $\sum_{y \in A} u_i(y) = 1$ imply $u_i(x) \leq 1$. Thus every alternative x satisfies

$$\text{SW}_{\mathbf{u}}(x) \leq n.$$

It follows that

$$\frac{\max_{x \in A} \text{SW}_{\mathbf{u}}(x)}{\text{SW}_{\mathbf{u}}(a)} \leq \frac{n}{\alpha n / m^2} = \frac{m^2}{\alpha}.$$

This bound is uniform over all profiles and all consistent normalized utility profiles. $\square$

5 Metric distortion

Fix a profile σ , set $a = f^\alpha(\sigma)$, and fix an arbitrary metric d consistent with the profile. By the definition of the rule, $D_H(a)$ has a perfect matching; fix a corresponding bijection $\pi : N \rightarrow N$, so that

$$a \succeq_i r_{\pi(i)} \quad \text{for every } i \in N.$$

Lemma 3. *For every comparison alternative $o \in A$,*

$$\text{SC}_d(a) \leq 2 \text{SC}_d(o) + R, \quad \text{where } R = \sum_{j \in N} d(j, r_j).$$

Proof. Metric consistency and the matching edge incident to voter i give

$$d(i, a) \leq d(i, r_{\pi(i)}).$$

Two applications of the triangle inequality, along the path $i, o, \pi(i), r_{\pi(i)}$, give

$$d(i, r_{\pi(i)}) \leq d(i, o) + d(o, \pi(i)) + d(\pi(i), r_{\pi(i)}).$$

Summing these inequalities over $i \in N$ yields

$$\begin{aligned} \text{SC}_d(a) &\leq \sum_{i \in N} d(i, r_{\pi(i)}) \\ &\leq \sum_{i \in N} d(i, o) + \sum_{i \in N} d(o, \pi(i)) + \sum_{i \in N} d(\pi(i), r_{\pi(i)}). \end{aligned}$$

Because π is a bijection and d is symmetric,

$$\sum_{i \in N} d(o, \pi(i)) = \sum_{j \in N} d(o, j) = \sum_{j \in N} d(j, o) = \text{SC}_d(o),$$

and likewise

$$\sum_{i \in N} d(\pi(i), r_{\pi(i)}) = \sum_{j \in N} d(j, r_j) = R.$$

The first sum is also $\text{SC}_d(o)$, which proves the claim. $\square$

It remains to bound R . Let t_j denote voter j 's original first-ranked alternative. Partition the voters into

$$G = \{j \in N : t_j \in H\}, \quad E = N \setminus G,$$

and write

$$g = |G|, \quad e = |E|.$$

Every voter in E has an original first choice outside H , and conversely every voter whose first choice lies outside H belongs to E . Consequently,

$$e = \sum_{b \in A \setminus H} p(b).$$

For every excluded alternative $b \notin H$, the definition of H gives the strict inequality

$$p(b) < \frac{\alpha n}{m}.$$

If $A \setminus H$ is nonempty, summing this strict inequality gives

$$e < |A \setminus H| \frac{\alpha n}{m} \leq \alpha n.$$

If $A \setminus H$ is empty, then $e = 0 < \alpha n$ directly. Thus, in all cases,

$$e < \alpha n, \quad g = n - e > (1 - \alpha)n > 0.$$

In particular, G is nonempty.

Lemma 4. *For every comparison alternative $o \in A$,*

$$R \leq \left(1 + \frac{2\alpha}{1 - \alpha}\right) \text{SC}_d(o).$$

Proof. Because G is a nonempty finite set, we may choose $k \in G$ minimizing $d(k, o)$. The minimum is no larger than the average, so

$$d(k, o) \leq \frac{1}{g} \sum_{i \in G} d(i, o) \leq \frac{\text{SC}_d(o)}{g}.$$

Since $k \in G$, its original first choice t_k belongs to H .

If $j \in G$, then $t_j \in H$. Since t_j is voter j 's first choice among all alternatives, it is also their highest-ranked member of H ; hence

$$r_j = t_j.$$

Moreover, $t_j \succeq_j o$, so metric consistency gives

$$d(j, r_j) = d(j, t_j) \leq d(j, o).$$

Now suppose $j \in E$. The alternative r_j is voter j 's highest-ranked member of H , and $t_k \in H$. Therefore

$$r_j \succeq_j t_k,$$

and metric consistency implies

$$d(j, r_j) \leq d(j, t_k).$$

The triangle inequality along the path j, o, k, t_k gives

$$d(j, t_k) \leq d(j, o) + d(o, k) + d(k, t_k).$$

Because t_k is voter k 's original first choice, $t_k \succeq_k o$. Metric consistency and symmetry therefore give

$$d(k, t_k) \leq d(k, o) \quad \text{and} \quad d(o, k) = d(k, o).$$

Combining these inequalities, for every $j \in E$,

$$d(j, r_j) \leq d(j, o) + 2d(k, o).$$

We may now sum separately over G and E :

$$\begin{aligned} R &= \sum_{j \in G} d(j, r_j) + \sum_{j \in E} d(j, r_j) \\ &\leq \sum_{j \in G} d(j, o) + \sum_{j \in E} (d(j, o) + 2d(k, o)) \\ &= \text{SC}_d(o) + 2e d(k, o) \\ &\leq \left(1 + \frac{2e}{g}\right) \text{SC}_d(o). \end{aligned}$$

Finally, $e < \alpha n$ and $g > (1 - \alpha)n > 0$ imply

$$\frac{e}{g} < \frac{\alpha n}{(1 - \alpha)n} = \frac{\alpha}{1 - \alpha}.$$

Replacing this strict inequality by the corresponding weak upper bound gives

$$R \leq \left(1 + \frac{2\alpha}{1 - \alpha}\right) \text{SC}_d(o). \quad \square$$

Combining Lemmas 3 and 4, for every $o \in A$ we obtain

$$\begin{aligned} \text{SC}_d(a) &\leq 2 \text{SC}_d(o) + \left(1 + \frac{2\alpha}{1 - \alpha}\right) \text{SC}_d(o) \\ &= \left(3 + \frac{2\alpha}{1 - \alpha}\right) \text{SC}_d(o). \end{aligned}$$

Since A is finite, it contains a social-cost minimizer o^* . Taking $o = o^*$ gives

$$\text{SC}_d(a) \leq \left(3 + \frac{2\alpha}{1 - \alpha}\right) \min_{x \in A} \text{SC}_d(x).$$

This is an undivided inequality. In particular, if the optimal social cost is zero, it forces $\text{SC}_d(a) = 0$, so the zero-cost convention in the definition of metric distortion is respected. The bound is uniform over all profiles and all consistent metrics. Together with Lemma 2, this proves Theorem 1.

6 Consequences and solution status

Corollary 1. *Taking $\alpha = 1/3$ gives a deterministic full-ranking rule satisfying*

$$\text{dist}_M(f^{1/3}) \leq 4, \quad \text{dist}_U(f^{1/3}) \leq 3m^2.$$

Consequently, for every $\delta > 0$,

$$\text{dist}_M(f^{1/3}) \leq 4 < 9 + \delta,$$

while the utilitarian distortion remains $O(m^2)$, with constant 3 independent of m , n , and δ .

Proof. Substitution gives

$$3 + \frac{2(1/3)}{1 - 1/3} = 3 + \frac{2/3}{2/3} = 4$$

and

$$\frac{m^2}{1/3} = 3m^2.$$

The inequality $4 < 9 + \delta$ holds for every $\delta > 0$. □

Corollary 2. *For every fixed $\varepsilon > 0$, there is a deterministic full-ranking rule satisfying*

$$\text{dist}_M(f) \leq 3 + \varepsilon, \quad \text{dist}_U(f) \leq \left(1 + \frac{2}{\varepsilon}\right) m^2.$$

Proof. Set

$$\alpha = \frac{\varepsilon}{2 + \varepsilon}.$$

Because $\varepsilon > 0$, this parameter satisfies $0 < \alpha < 1$, and

$$1 - \alpha = \frac{2}{2 + \varepsilon}.$$

Hence

$$\frac{2\alpha}{1 - \alpha} = \frac{2\varepsilon/(2 + \varepsilon)}{2/(2 + \varepsilon)} = \varepsilon,$$

while

$$\frac{1}{\alpha} = \frac{2 + \varepsilon}{\varepsilon} = 1 + \frac{2}{\varepsilon}.$$

The conclusion follows from Theorem 1. □

Corollary 1 completely resolves the target existence problem: no gap remains concerning the requested $9 + \delta$ metric bound and the $O(m^2)$ utilitarian bound. Corollary 2 is stronger in that the metric constant can be made arbitrarily close to 3 while retaining quadratic dependence on m ; it does not assert the endpoint value 3. The parameterized rules vary with ε , and the utilitarian coefficient $1 + 2/\varepsilon$ diverges as $\varepsilon \downarrow 0$.

7 Boundary cases

The construction and proofs cover the following boundary cases.

- If $H = A$, then $E = \emptyset$, so $e = 0$. Every r_j is voter j 's original first choice, and the proof gives

$$R \leq \text{SC}_d(o), \quad \text{SC}_d(a) \leq 3 \text{SC}_d(o).$$

- If H is a singleton, say $H = \{h\}$, then $r_j = h$ for every voter j . The graph $D_H(h)$ is complete, because for every pair of voters i, j the edge condition reads $h \succeq_i h$, which always holds. Every bijection $N \rightarrow N$ is therefore an explicit perfect matching, and the rule selects h .
- An alternative whose plurality score is exactly $\alpha n/m$ is retained. Thus every excluded alternative satisfies the strict inequality $p(b) < \alpha n/m$, which is exactly what yields $e < \alpha n$.

- If the optimal social cost is zero, the proved undivided approximation inequality forces the selected social cost to be zero as well. Hence the argument never divides by zero.
- If $m = 1$, the unique alternative belongs to H , every graph used in the construction is complete, and the rule selects that alternative. All displayed bounds remain valid for every $n \geq 1$.

No external matching theorem or other specialized result is required: Lemma 1 constructs the matching directly, and all remaining steps use only finite averaging, consistency of the reported rankings with utilities or distances, bijectivity of the constructed matching, and the triangle inequality.

References

- [1] Vasilis Gkatzelis, Mohamad Latifian, Nisarg Shah. *Best of Both Distortion Worlds*. Economics and Computation (EC), 2023. <https://doi.org/10.1145/3580507.3597739>